

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 3C

Revision and application: choosing a paper supplier

Bellerbys
Preparation for University Education

Name _____ Class _____ Date _____

REVISION

Classes A and B

Eight trial orders help us estimate a supplier's overall delivery performance. **This warm-up uses separate data from the main case.**

Delivery times (h): 40, 43, 44, 46, 49, 51, 54, 57

$$\bar{x} = 48 \text{ h}, \quad \sum (x_i - \bar{x})^2 = 236 \text{ h}^2, \quad Q_1 = 43.25 \text{ h}, \quad Q_3 = 53.25 \text{ h}$$

Calculator allowed. Show working and units. Keep the unrounded SD for CV.

1. Sample or population?

Explain your choice. Give the symbols for its mean and standard deviation (SD).

2. Find the median and range.

With an extreme delay, which changes less: mean or median? Explain briefly.

3. Find the interquartile range (IQR).

Show the calculation. Which part of the data does IQR describe?

4. Find the deviation for 57 hours.

Show the calculation. Explain how a deviation is different from SD.

5. Calculate sample variance and SD.

Show both calculations. Explain your choice of divisor.

6. Find the coefficient of variation (CV).

Show the calculation. What would a smaller CV mean?

THE CASE: CHOOSE A PAPER SUPPLIER

A university print shop needs a regular A4-paper supplier. Both offer the **same quality, quantity and total price, including delivery**. Routine orders can be placed a few hours earlier. Use the separate sheet: **30 simulated trial orders per supplier**. Times run from confirmation to arrival, include nights and weekends, and are recorded to the nearest hour.

EXERCISE C1 Average delivery time

Add all 30 times for each supplier. Use the subtotals to check your addition.

Working / result	Supplier A	Supplier B
Trials 1-10: subtotal (h)		
Trials 11-20: subtotal (h)		
Trials 21-30: subtotal (h)		
Total of all 30 (h)		
Mean: total ÷ 30 (h)		

$$\bar{x} = \frac{\sum x_i}{n}$$

$\bar{x}$ (**x-bar**): sample mean
 Σ (**capital sigma**): add
 x_i : one time; n : count (30)

Which is faster on average, and by how many hours?

EXERCISE C2 Median, range and quartiles

1. Sort both sets. Write one delivery time in each blank cell.

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A															
B															

Position	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A															
B															

$$Q_1 \text{ position: } \frac{30+1}{4} = 7.75 \quad Q_3 \text{ position: } \frac{3(30+1)}{4} = 23.25$$

Median: average the 15th and 16th values. Range: largest minus smallest.

Q1: 75% of the gap from the 7th to 8th value. Q3: 25% from the 23rd to 24th.

Measure	Supplier A: working + answer (h)	Supplier B: working + answer (h)
Median		
Range		
Lower quartile, Q1		
Upper quartile, Q3		
IQR = Q3 - Q1		

2. Does an equal range mean equally consistent deliveries? Use your IQR results.

VARIATION: SUPPLIER A

EXERCISE C3 Supplier A: all 30 deliveries

Mean = 46.8 h. A deviation is one delivery time minus this mean. Then square the deviation. Check your work after each ten rows.

The first row is worked for you. A negative deviation means a delivery below the mean.

Trial	Delivery time, x_i (h)	Deviation: $x_i - 46.8$ (h)	Squared deviation (h^2)
1	36	$36 - 46.8 = -10.8$	116.64
2	49		
3	45		
4	54		
5	41		
6	31		
7	43		
8	38		
9	68		
10	46		
11	50		
12	65		
13	40		
14	39		
15	45		
16	60		
17	52		
18	35		
19	44		
20	42		
21	51		
22	47		
23	33		
24	62		
25	55		
26	43		
27	48		
28	28		
29	56		
30	58		
	TOTAL		

Add the squared-deviation column in the final cell. This total is not the variance yet.

VARIATION: SUPPLIER B

EXERCISE C3 Supplier B: all 30 deliveries

Mean = 49.2 h. A deviation is one delivery time minus this mean. Then square the deviation. Check your work after each ten rows.

Complete the first row with your teacher, then continue independently. Keep both tables.

Trial	Delivery time, x_i (h)	Deviation: $x_i - 49.2$ (h)	Squared deviation (h^2)
1	51		
2	44		
3	49		
4	47		
5	48		
6	49		
7	51		
8	28		
9	68		
10	52		
11	56		
12	49		
13	46		
14	52		
15	50		
16	55		
17	47		
18	45		
19	50		
20	47		
21	53		
22	48		
23	48		
24	46		
25	54		
26	45		
27	50		
28	53		
29	44		
30	51		
	TOTAL		

Add the squared-deviation column in the final cell. This total is not the variance yet.

FROM SQUARED DEVIATIONS TO A COMPARISON

EXERCISE C3 Sample variance and standard deviation

The 30 trial orders are a **sample** used to estimate wider delivery performance. For each supplier, $n = 30$, so the sample variance divisor is $n - 1 = 29$.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \qquad s = \sqrt{s^2}$$

s^2 (**s squared**) = sample variance, in h^2 . **s** = sample standard deviation, in hours.

Calculation	Supplier A: working and answer	Supplier B: working and answer
Squared-deviation total (h^2)		
Sample variance (h^2) Total $\div 29$		
Sample SD (h) Square root of variance		

Keep the full SD in your calculator for C4. Round final variance and SD to 2 decimal places.

EXERCISE C4 Coefficient of variation (CV)

CV expresses SD as a percentage of the mean. Round the final CV to 2 decimal places.

$$CV = \frac{s}{\bar{x}} \times 100\%$$

Supplier A: unrounded SD $\div$ mean $\times 100\%$

Supplier B: unrounded SD $\div$ mean $\times 100\%$

Evidence for your recommendation. Copy your checked results. Include units.

Measure	Supplier A	Supplier B
Mean (h)		
Median (h)		
Interquartile range (h)		
Sample standard deviation (h)		
Coefficient of variation (%)		
Longest recorded delivery (h)		

A RECOMMENDATION FOR ROUTINE ORDERS**EXERCISE C5****Your recommendation**

Choose a supplier for routine orders. The print shop can order a few hours earlier. Use complete sentences, compare the figures, and explain what they mean for the business.

1. For routine orders, I recommend...

State your choice clearly.

2. The delivery records support this because...

Compare both means and at least one measure of spread (SD or IQR). Include hours.

3. For this print shop, this matters because...

Explain how the evidence affects ordering or avoiding interruptions.

4. The other supplier's main advantage is...

Acknowledge the trade-off and one remaining delivery risk.

Check your response: a clear choice, numerical comparison, business meaning, a trade-off and a remaining risk.

One improvement after feedback:

STOP. Keep the next page closed until your teacher introduces the new case.

A NEW REQUIREMENT: THE URGENT ORDER**EXERCISE C6a The 48-hour deadline**

The shop must order **now** for an urgent job. The paper must arrive **within 48 hours**. Count recorded times of **48 hours or less (≤ 48)**.

$$\text{Percentage on time} = \frac{\text{deliveries within deadline}}{30} \times 100\%$$

Evidence from the 30 trials	Supplier A	Supplier B
Deliveries within 48 h: count		
On-time percentage: working and result		
Deliveries over 48 h: count		

1. For an order needed within 48 hours, I would...

State your recommendation.

2. In the trial records...

Compare both suppliers' on-time counts and percentages.

3. My choice still has a risk because...

Explain what could go wrong, using the delivery evidence.

4. The print shop could reduce this risk by...

Suggest a practical action for this urgent order.

These percentages describe the trial records. They do not guarantee the next delivery.

STOP. Wait for your teacher before turning to the revised deadline.

A REVISED DEADLINE AND YOUR FINAL EXPLANATION

EXERCISE C6b Six more hours

The job starts six hours later. The order time stays the same. Count recorded times of **54 hours or less (≤ 54)**, then reconsider your recommendation.

New evidence	Supplier A	Supplier B
Deliveries within 54 h: count		
On-time percentage: working and result		
Extra deliveries included since 48 h		

1. With a 54-hour deadline, my recommendation is...

State your decision for the new deadline.

2. Compared with the 48-hour deadline, the records now show...

Use both suppliers' percentages and explain the comparison.

3. My recommendation changes / stays the same because...

Explain why the extra six hours matter.

EXERCISE C7 Individual exit question

A manager says: **"Always choose the supplier with the smallest SD."** Do you agree? Use at least one numerical comparison and explain it.

1. A smaller standard deviation tells us...

Explain what the statistic measures.

2. For a particular deadline, we also need to consider...

Connect the statistic to the business requirement.

3. Our delivery records illustrate this because...

Include numerical evidence and explain what it shows.